

the Universe could have started from a singularity. He compared the beginning of the Universe to the South Pole, saying: "To ask what happened before the beginning of the Universe would become a meaningless question, because there is nothing south of the South Pole."

Hawking radiation

Following a debate with Israeli physicist Jacob Bekenstein in 1972, Hawking refined his black hole theories to take into account the laws of quantum mechanics, which deals with the behavior of matter and energy on the scale of atoms and subatomic particles. These laws state that throughout the Universe, pairs of particles and antiparticles are constantly meeting and annihilating each other. Hawking calculated that when this occurs near the event horizon (the visible edge of a black hole), the negatively charged particle of the pair falls into the black hole and

the positively charged half of the pair streams away from the black hole's edge as radiation—which became known as Hawking radiation. According to Hawking, the black hole loses energy and mass in a process called black hole evaporation as the negatively charged particles fall into it. Over eons, the black hole continues to leak radiation and particles until it disappears in a final explosion.

In 1974, Hawking published his calculation showing both how the Universe began and how black holes evaporate. By combining insights from the theory of the infinitely large (Einstein's general theory of relativity)

ROGER PENROSE

British mathematician and theoretical physicist
Roger Penrose is known for his work on black holes and the Big Bang theory.



In the 1960s, Penrose (1931–) was a reader and then professor in applied mathematics at Birkbeck College, London, where he developed his work on black hole singularities. In 1969, he collaborated with Stephen Hawking to prove that black holes can arise from the gravitational collapse of massive stars, and their paper was published by the Royal Society of London in 1970. Penrose also developed twistor theory, which is a key tool in quantum theory, and created a way of mapping the regions of space-time around a black hole. Featuring so-called Penrose diagrams, this mapping technique makes it possible to visualize the effect that gravity has on an object approaching a black hole.

Hawking's discovery that black holes are not really black, but rather emit thermal radiation due to quantum effects, was one of the most important findings of 20th-century physics. Initially, his formula for Hawking radiation was rejected by many of his peers, but it has since become universally accepted.